

Constantinos Petrides

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EDUCATION

•BSc. Physics - Environmental Physics and Meteorology

National and Kapodistrian University of Athens, Athens, Greece

Thesis: Air pollution in the region of Attica during the COVID-19 Lockdown

•Cypriot Apolitirion (High School Diploma)

Kykos B High School, Nicosia, Cyprus

Courses: Programming, Advanced Mathematics, Advanced Physics, Greek, Italian, Spanish

PROFESSIONAL EXPERIENCE

•Research Intern

2026

The Cyprus Institute, Cyprus

- Conducted a ground-level intercomparison of UAV-based aerosol sensors.
- Member of Prof. Franco Marengo's research group.

•Research Assistant

2024 – 2026

National and Kapodistrian University of Athens, Greece

- Contributed to FAIRMODE bias correction methodologies for air quality modeling.
- Developed Python tools for atmospheric data analysis and visualization.

RESEARCH EXPERIENCE

•Bias Correction of Air Quality Models

FAIRMODE WG5 (European Commission JRC & EEA), via NKUA, Athens, Greece

2025

- Implemented bias correction techniques on air quality model outputs ($\text{PM}_{2.5}$, NO_2 , O_3) using spatial interpolation methods and machine learning techniques.
- **Authors:** FAIRMODE WG5 (NKUA contributors: E. Bossioli, C. Petrides)
- **Status:** Ongoing project; initial analysis available at [FAIRMODE](#)

•A Hybrid Quantum–Classical Solver for the Incompressible Navier–Stokes Equations

GitHub repository

2025

- Implemented a solver for the incompressible Navier-Stokes equations using the classical SIMPLE method and a hybrid classical-quantum approach for pressure correction. Demonstrated integration of quantum algorithms (VQE) within computational fluid dynamics.
- **Authors:** C. Petrides
- **Repository:** [GitHub](#)

•NO_x–Ozone Relationship Modeling Using a Machine Learning Approach

2024

GitHub repository

- Developed models to analyze and predict the nonlinear relationship between NO_x (nitrogen oxides) and ozone concentrations using methods such as regression analysis, support vector machines, random forests, and neural networks, effectively capturing complex dependencies in environmental data.
- **Authors:** C. Petrides
- **Repository:** [GitHub](#)

•Numerical Simulation of the 2D Shallow Water Equations

2023

GitHub repository

- A shallow water simulation designed to model fluid dynamics, visualize surface elevations and velocity fields, and analyze key parameters such as the Rossby radius and wave propagation speeds under various initial conditions.
- **Author:** C. Petrides
- **Repository:** [GitHub](#)

•Time Series Forecasting of Meteorological Data Using Machine Learning and Statistical Models

2023

GitHub repository

- Developed and evaluated forecasting models (ETS, Prophet, LSTM, GRU, Random Forest) to predict temperature trends from historical weather data. Applied feature engineering and model tuning to enhance accuracy, demonstrating skills in both statistical and deep learning time series analysis..
- **Author:** C. Petrides
- **Repository:** [GitHub](#)

CONFERENCES & SEMINARS

•Attendee, Current practices and challenges concerning emissions projections for planning applications 2026

FAIRMODE, European Commission, JRC

•Attendee, WG8 Webinar: Modeling and Spatial Representativeness

2026

FAIRMODE, European Commission, JRC

•Attendee, From Evidence to Impact: Strengthening Research, Policy, and Capacity Ecosystems

2025

UN SDSN, European Commission JRC, Maldives Bureau of Statistics

•Participant, 6th Pan-Cyprian Student-Teachers Conference in Natural Sciences, Nicosia, Cyprus

2017

Cyprus Ministry of Education, Sport and Youth

•Participant, 5th Pan-Cyprian Student-Teachers Conference in Natural Sciences, Nicosia, Cyprus

2016

Cyprus Ministry of Education, Sport and Youth

AWARDS

•Municipality of Kyrenia Award

2017

Municipality of Kyrenia Academic Achievement Award

•Academic Achievement Award

2017

Kykkos B High School, Nicosia, Cyprus

•Innovation Award for the presentation of “Whisper Dishes”

2017

6th Pan-Cyprian Student-Teachers Conference in Natural Sciences, Paphos, Cyprus

•Innovation Award for the presentation of “Solar Air Heater”

2016

5th Pan-Cyprian Student-Teachers Conference in Natural Sciences, Nicosia, Cyprus

TECHNICAL SKILLS

Languages: Greek (native), English (proficient), Spanish, Italian and BSL (elementary)

Developer Tools: Python, C++, SQL, R, Java, HTML, CSS